THE DRAWING(S) AND THE INFORMATION CONTAINED HEREIN ARE THE PROPERTY OF **WOOJIN INDUSTRIAL SYSTEMS CO., LTD.** THEY SHALL NOT BE DISCLOSED, REPRODUCED OR USED, IN ANY MANNER WITHOUT THE PRIOR WRITTEN CONSENT OF **WOOJIN INDUSTRIAL SYSTEMS CO., LTD.**

DENVER RTD EAGLE P3

Communication system

Software Configuration Management Plan

(SCMP)

DISTRIBUTION	
TOTAL	

REVISION							
No.	ISSUED DATE	REVISION NOTE	DESIGNED	CHECKED	APPROVED	REVIEWED	
			ENGINEERING			PM	

PM	REVIEWED	E.K. LEE
ENGINEERING	APPROVED	B.J. JONE
	CHECKED	D.S. JANG
	DESIGNED	D.B. LEE
ISSUED DATE	June 14, 2011	
ISSUED BY	R & D CENTER	
PROJECT NO.	P2011-020	
ORDER NO.	P110701149	
DOCUMENT NO.		REVISION
WJ-RTD-COMM-SCMP-V1.0		0

CUSTOMER :

HYUNDAI-ROTEM COMPANY

PROJECT :

COMMUNICATION SYSTEM FOR DENVER RTD EAGLE P3 PROJECT

WOOJIN Industrial Systems Co., Ltd.

REVISION RECORD

Rev. No.	PAGE	Details		Date
		FROM	TO	
0	All	Issued		June 14, 2011

Table of Contents

1. Overview 4
1.1. Scope 4
1.2. Reference Documents 4
1.3. Definitions and acronyms 5
2. SCM Management 7
2.1. Organization 7
2.2. SCM Roles 8
2.3. Tasks and Responsibility 9
2.4. Applicable policies, directives, and procedures 9
3. SCM activities 10
3.1. Configuration Identification 10
3.2. Configuration control 14
3.3. Configuration status accounting 16
3.4. Configuration audits and reviews 17
3.5. Interface control 18
4. SCM Schedules 18
5. SCM resources 19
6. SCM plan maintenance 19

1. Overview

1.1. Scope

This Software (called "SW" hereafter) Configuration Management Plan applies to the work-products including tools and are created through the entire life cycle of the software development process for the EMUs supplied to the Denver RTD (Regional Transportation District.

The items to be identified and managed by configuration management activities shall be deliverables from the tasks in software lifecycle phases, includes development related plans, requirements and design description and verification and validation reports.

This document shall be issued to and used by Configuration Manager, Quality Assurance Manager, Project Manager, and Design Team through all over the software development lifecycle.

The audience for this document consists of the following groups and individuals:

- WOOJIN's software engineering group, project management and software quality assurance
- Rotem and DENVER representatives

1.2. Reference Documents

This document shall be used in conjunction with the following documents and publications. If the reference documents or publications are superseded by an approved revision, the most recent revision shall apply.

- IEEE Standard 828, IEEE Standards for Software Configuration Management plans

1.3. Definitions and acronyms

1.3.1. Definitions

The definition for the below terms is described as follows:

Terms	Definitions
Version Management	The establishment and maintenance of baselines and the identification of changes to baselines that make it possible to return to the previous baseline.
Change Management	Judicious use of means to effect a change, or a proposed change, on a product or service.
CM: Configuration Management	A discipline applying technical and administrative direction and surveillance (1) to identify and document the functional and physical characteristics of a configuration item, (2) to control changes to those characteristics, (3) to record and report change processing and implementation status, and (4) to verify compliance with specified requirements.
SCI: software Configuration Items	An aggregation of work products that is designated for configuration management and treated as a single entity in the configuration management process. The SCI is distributed, installed and version tracked as one item.
Baseline	1) The configuration information formally designated at a specific time during a product's or product component's life. Configuration baselines, plus approved changes from those baselines, constitute the current configuration information. 2) A document or product that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures.
Configuration Library	Physical, Logical storage space that saving configuration items. Ex) File server, configuration management system by tool.
CCB: Configuration Control Board	A group of people responsible for evaluating and approving or disapproving proposed changes to configuration items, and for ensuring implementation of approved changes.
Configuration Identification	An element of configuration management consisting of selecting the configuration items for a product, assigning unique identifiers to them, and recording their functional and physical characteristics in technical documentation.
Configuration Control	An element of configuration management consisting of the evaluation, coordination, approval or disapproval, and implementation of changes to configuration items after formal establishment of their configuration identification.
Configuration Status Accounting	An element of configuration management consisting of the recording and reporting of information needed to manage a configuration effectively. This information includes a listing of the approved configuration identification, the status of proposed changes to the configuration, and the implementation status of approved changes.
Configuration Audits	An audit conducted to verify that a configuration item, or a collection of configuration items that make up a baseline, conforms to a specified standard or requirement.

Terms	Definitions
Check-Out	For change configuration item passed baseline, bring configuration item needed to change from configuration library to requestor's library or test library.
Check-In	Register initial configuration item passed baseline at configuration library or complete change activity for configuration item and re-register the configuration item at configuration library.
Release	The formal notification and distribution of an approved version.
Commercial – Off- The – shelf	General market software items typically supplied to a wide range of users for a range of applications.
Validation	Does the product meet the system requirements? (Are we building the right product?)
Verification	Does the product match the design? (Are we building the product right?)

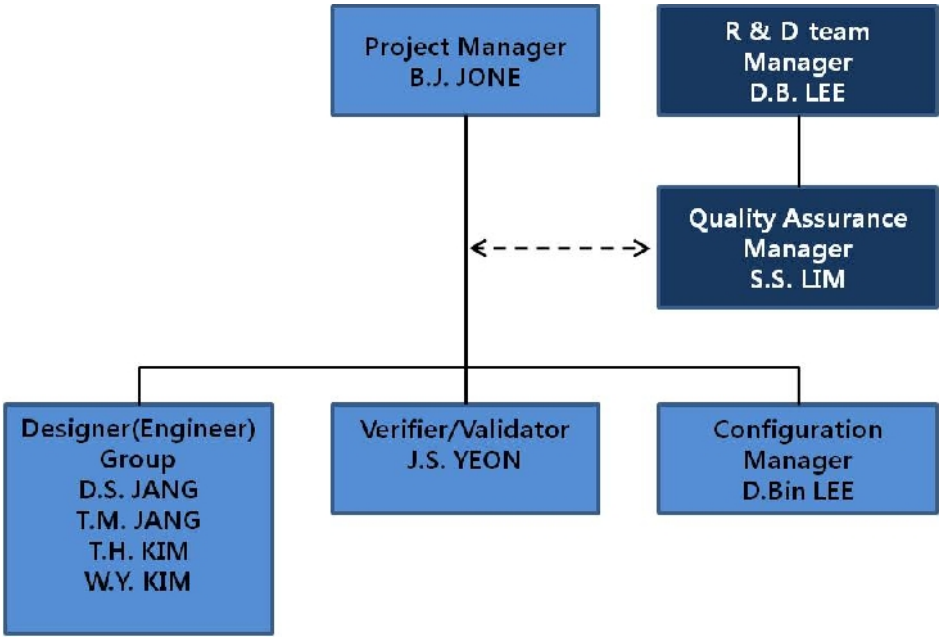
1.3.2. Acronyms

Acronyms	Contents
CCB	Configuration control board
CI	Configuration item
SCM	Software configuration management

2. SCM Management

2.1. Organization

Organizational unit for SCM Management activities and organizational unit for CCB are as follows:



2.2. SCM Roles

Roles	Description
Project Manager	● Support and manage the SW Configuration Management activities for the project. ● Review and approve the SW Configuration Management Plan. ● Consist of the SW Configuration Management Organizational units units (SCM officer, CCB) for the project. ● Perform CCB's chairman roles for the project and has authorities of final approval regarding to all configuration items and changes request. ● Review the SCM activities and results for the project regularly. ● Report the agenda of Project SCM activities to management
S/W Configuration Manager	● Establish, distribute and maintenance the SCM plan. ● Build up the SCM system, provide the access authority, and manage back-up. ● Train the SW configuration management for the project team. ● Select, identifies and review the SW configuration items. ● Request the baseline audit to CCB. ● Set-up, change management and notify the configuration baseline. ● Distribute product from configuration baseline. ● Accept and manage the request form of changes for configuration items. ● Record and report the SCM activities, configuration status. ● Perform the configuration audit instead of QA manager if necessary.
CCB (DENVER, ROTEM, WOOJIN)	● Admit the configuration baseline set-up. ● Perform the effect analysis, evaluate, and audit for the changes requests caused from customer, software engineering group.
Quality Assurance Manager	● Review the SCM plan. ● Perform the inspection for the quality regarding to SCM activities and work-products. ● Perform the configuration audit.
Software Engineering Group	● Develop the configuration items by staged and request the changes of configuration items if necessary. ● Establish the changes for the configuration item. ● Perform the corrective action of nonconformance caused configuration audit results.
Customer	● Request the changes for established configuration baseline. ● If critical changes influencing entire project are generated, participate in CCB and then evaluate and approve the change's effect according to project manager's request.
Verifier/Validator	● Verification and Validation activities of the project work products ● Author and maintain test plan and test report ● Control test records and track defects

2.3. Tasks and Responsibility

☞'∨' means 'or'

Task	CCB	Quality Assurance	Designer	Project Manager	Configuration Manager	Customer (RTD, ROTEM)
Configuration Identification	Review	Audit	Review	✓ Originate Approve		Approve
Configuration Control	Approve	✓ Approve Audit	✓ Request ✓ Evaluate Implement	✓ Request ✓ Evaluate ✓ Approve Implement	Implement	✓ Request ✓ Evaluate Approve
Configuration Status Accounting		Audit	✓ Originate ✓ Review Implement	✓ Originate ✓ Review ✓ Approve Implement	✓ Originate Implement	
Configuration Audits & Reviews		Audit	Review	✓ Audit Review	Audit	Review

2.4. Applicable policies, directives, and procedures

The policies and procedures described in the project Software Quality Assurance Plan section

- 3.4 Software Development Lifecycle Phases, Tasks and Deliverable
- 3.7 Documents Control and Revisions
- 10.0 Media Control
- 12.0 Quality Records

are the fundamental to software configuration management.

3. SCM activities

SCM activities information identifies all functions and tasks required to manage the configuration of the software system as specified in the scope of the SCMP. Both technical and managerial SCM activities shall be identified. SCM activities grouped into four functions (configuration identification, configuration control, status accounting, and configuration audits and reviews) traditionally are identified below 3.1 through 3.4.

3.1. Configuration Identification

This SCMP mentions the project configuration items and their structures at each project control point, mentions how each CI and its version are to be uniquely named, and mentions the activities to be performed to define, track, store, and retrieve SCIs.

3.1.1. Identifying configuration items

Configuration identification shall be applied to both source code and documentation.

- SCIs, including deliverable items and support tools
- Source code used to build the SCIs, together with any COTS libraries and identification of development tools used
- Software Plans (SQAP, SCMP, SVVP, SPMP)
- System and software requirements specifications (SRS)
- Interface Control Document (ICD)
- System Functional Description (SFD)
- System Design Description (SDD)
- Software Test Plan
- Software Test Procedures
- Software Test Report
- Software Verification and Validation Report
- Software Requirement Traceability Metrics
- Software user documentation

3.1.2. Tool

Phase	Software Tool	Version information
Implement	DSP : AMC Emulator	1.5.0.D
	DSP: , UltraEditor	11.10a
	ATMEGA : AVRSTUDIO	4.17
	WINDOWS : VISUAL STUDIO 2008	9.0

3.1.3. Naming configuration Items

This SCMP mentions an identification system for assigning unique identifiers to each item to be controlled. Identification methods could include naming conventions and version numbers and letters.

A unique number is assigned to each formal document generated. Each time the document is updated the revision letter is incremented and a description of the change is logged on the revision page.

(1) Document

[1] Company Name – [2] Project Name – [3] System Name - [4] Document name, [5] R. [6] Extension

1) Company: Company Name (WJ: Woojin Industrial Systems Co., Ltd)
2) Project: Project Name (RTD: Regional Transportation District)
3) system: System Name (COMM: Communication System)
4) Document Name: Document abbreviation
   - Software Configuration Management Plan: SCMP
   - Software Project Management Plan: SPMP
   - Software Configuration Management Plan: SQAP
   - Software Verification and Validation Report: SVVR
   - Software Design Description: SDD
   - Software Requirement Specification: SRS
   - Software Functional Description: SFD
   - Software Requirement Traceability Matrix: SRTM
   - Software Source Program: SC
   - Software Test Plan: STP
   - Software Test Procedure: STPr
   - Software Test Report: STR
5) Rev.: Revision Number (change Requirement Counting: +0.1)
6) Extension: In accordance with Program

(2) Source Program

1) Project – 2) System – 3) Device – 4) Program – 5) Ver. 6) Extention

1) Project: Name of Project
2) System: Name of System
3) Device: Name of Device
4) Program: Name of Program

5) Ver : Version
①Revision Code ②0. ③00

① Revision Code: A ~ Z (② number is over 9 digit.)

② Base Line starts counting

③ Change Requirement: + 0.01

6) Extension : In accordance with Coding program

3.1.4. Implementation configuration Management System

The main rule is a staff authorized by project manager shall be able to have the ability to update a controlled item at any one time. The library system shall "check out" an item for update, and "check in" the item when the work has been completed, checked and approved for update. For access and authorization rules that should be applied to ensure people follow the procedures, refer to the figure right below.

Management Level	Location	Development group	SCM Officer	PM
Version Management	Develop Library	RW	RW	RW
Baseline Management	Master Library	R	RW	R

3.1.5. Release Procedure

3.1.5.1. Release Type

Throughout the whole phases of the development life cycle, at the discretion of the Configuration Manager Controls, baseline releases may be performed. The releases fall into one of three categories:

- Patch Release: Debegging Software release
- Preliminary Release: Software code comment and Software testing release
- Full Release: Approved Document, Software, Drawing release

3.1.5.2. Release Strategy

When a baseline is approved and changed, configuration manager makes a new baseline of master Library. Configuration manager must account baseline to project member after integrity check of configuration items. Customer release baseline make an entry information (label name, release name, release date, release time, release officer, etc) to release library.
The access control of release library is permitted to the configuration manager or specially assured member. After setting up baseline, all the changes of the baseline follow the configuration control procedure.
Engineering group officer reviews release baseline status, if the integrity is not assured, he makes new change request and the configuration control procedure shall be applied.

3.2. Configuration control

Configuration control activities request, evaluate, approve or disapprove, and implement changes to baselined CIs. Changes include all of the error correction, enhancement, and the control of interfaces/subcontracts/Vendors. The degree of formality necessary for the change process depends on the project baseline affected and on the impact of the change within the configuration structure.

3.2.1. Requesting changes

When baselined configuration items needs to be changed, project member or customer request for change with configuration items to be changed to the SCM officer.

3.2.2. Evaluating changes

Change requestor evaluates the criticality as basis of below classification for the changes.

Criticality	Criteria
High	Changes are affected to the range of project. ● Changes are affected to required performance, schedule, resources caused by requirement change ● Changes are needed to the contract changes (Range of project, contents, contract cost etc.) ● In case that the change of the configuration item affects to the other work-products
Medium	Changes that needs to be discussed with other related tasks, except for the criticality “high”. ● Changes that affect to compatibility and alternation ● Changes that affect to standards and procedures ● Changes of common module or common component
Low	Changes that can be handled within unit tasks, except for the criticality “high”, “medium”. ● Changes like typographical errors, mistake etc that affect little to the others

SC Manager accepts the request for configuration changes, tracks the configuration items connected with configuration items that should be changed, and then reviews the contents of the request.

SCM officer evaluates the criticality caused by the change request, if it is difficult to be judged, SCM officer requests the technical resources to development group in order to assess the impacts of change.

SCM officer calls CCB based on the following classification criteria.

Criticality	CCB member	CCB's roles
High	Customer PM SCM officer QA manager related persons	1st level CCB ● In case of criticality "high", determine 'approved', 'rejected', or 'held' through impact analysis and evaluating. ● Shall analyze cost of changes
Medium	PM SCM Officer related persons (when necessary, customer)	2nd level CCB ● In case of criticality "medium", determine 'approved', 'rejected', or 'held' through impact analysis and evaluating. ● In case of criticality "High" (determined by CCB review), submit that request to 1st level CCB.
Low	SCM Officer related persons (when necessary, PM)	3rd level CCB ● In case of criticality "Low", determine 'approved', 'rejected', or 'held' through impact analysis and evaluating. ● In case of criticality "Medium" (determined by CCB review), submit that request to 2nd level CCB.

3.2.3. Approvement or disapprovement of changes

1) CCB analyze the range and effect of changes based on the technical report and the change request, evaluate the impact that affects project, and determine 'approved', 'rejected', 'held'.
2) CCB record the result of effect analysis and evaluation of configuration changes. SCM officer participate in CCB as a writer and record the results of CCB meeting.
3) If approved, SCM officer notify the results and instructions from CCB to development group who is responsible for the changes of configuration items
4) If disapproved, that request process is finished, SCM officer notify the results from CCB to development group and keeps the related records.

3.2.4. Implementing changes

1) SCM officer identifies the baseline and the configuration items to be changed, and check-out that items from the configuration management system.
2) PM changes the project plan, if schedule and cost are changed.
3) If approved changes are different with the contract, PM discuss with the customer and then modify contents of the contract.
4) Development group who is responsible for performing the changes to configuration items, record the history of changes.
5) Development group and PM review and test the changed configuration items and confirm that the changes are successful. If re-work is needed as a result of review and test, person in charge perform that re-work.
6) SCM officer confirm whether all the related configuration items are changed, get approval of change completion from CCB.
7) SCM officer check-in the confirmed configuration items into configuration management system, and record the history of changes.
8) SCM officer notify the changes of configuration items and baseline to development group.

3.3. Configuration status accounting

For documents to be tracked and reported for baseline/change control, refer to section 3.1.1.

Configuration status accounting activities records and reports the status (change result, change effect, changing status) of project configuration items.

Configuration status accounting activities records and reports the status (change result, change effect, changing status) of project SCIs.

The following minimum data elements shall be tracked and reported for each configuration items

3.3.1. Configuration items status records

1) Recording time: all the phase of project life cycle.
2) SCM officer manage the status of configuration items continually and maintain the records.

3.3.2. Configuration status accounts

1) Reporting time: regularly(end of phases) according to the agreement or when change is occurred
2) SCM officer record the configuration status and report it to PM regularly/non-regularly.
   - CM-FM-001: Configuration Item Management Document (Configuration Revision History)
   - CM-CH-001: Configuration audit Check List (Function/ Physical)
   - CM-FM-002: Change Assessment / Request Foam
   - CM-FM-003: SW Revision History

3.4. Configuration audits and reviews

Configuration audits determine to what extent an actual configuration items reflects the required characteristics.

The QA Group is responsible for conducting QA audits to verify adherence to this SCM Plan. The CCB performs configuration reviews at product baseline release.

For deliverables to be audited, refer to section 3.1.1.

3.4.1. Preparing configuration audits

1) SCM officer check the integrity of configuration items prior to the configuration audits and then check the configuration management records and reports.
2) SCM officer notify the schedule of configuration audits to development group.

3.4.2. Implementing configuration audits

1) Configuration auditor: QA manager, SCM officer.
2) Configuration audit time: in the QA activities by phases.
3) QA manager direct the corrective action for the nonconformance to SCM officer as a result of configuration audits.
4) SCM officer direct the corrective action for nonconformance to development group as result of configuration audits.

3.5. Interface control

All interface control documents (ICD) are controlled as described in Configuration Management for Work Products. A document is placed under control of configuration management for purpose of release and distribution.

4. SCM Schedules

Phase	Beginning	Complete	2011								2012			
			5	6	7	8	9	10	11	12	1	2	3	4
Plan	11/05/01	11/06/15												
Requirements	11/06/16	11/07/31												
Design	11/08/01	11/08/31												
Implementation	11/09/01	11/12/31												
Test	12/01/01	12/02/28												
Maintenance	12/03/01													

- Major Milestone
  - 2011.05.01: Project Beginning
  - 2011.07.31: Requirements Complete
  - 2011.08.31: Design Complete
  - 2011.12.31: Product Complete
  - 2012.02.28: Test Complete
  - 2012.03.01: Maintenance Beginning

5. SCM resources

SCM resource consists of the software tools, techniques, equipment, personnel, and training necessary for the implementation of the specified SCM activities

Items	Resource/Role	Description
Tool	Configuration Management Server	Configuration Management Supporting Tool
	Software Source Coding Tool	AMC, AVRSTUDIO, VISUAL STUDIO
Techniques	Configuration Management Procedure	During all phases of the Project Life Cycle, we follow up the configuration identification, configuration control, configuration status account, configuration audit procedure.
Personnel	Software Configuration Control Board(CCB) Software Engineer	Refer to Section 2.1 organization and section 2.2 SCM responsibilities, they follow up SCM activities

6. SCM plan maintenance

To maintain SCMP, following items shall be monitored.

a) Who is responsible for monitoring that the plan (is implemented),
b) How frequently SCMP updated,
c) How changes to the plan are evaluated and approved, and
d) How changes to the plan are made and communicated.

The project Lead Software Engineer is responsible for monitoring and revising this Plan.

This Plan shall be updated whenever there is a change in the activities planned or in other information that it contains. As a minimum, this plan shall be revisited at each project development phase milestone defined in the SQAP to determine whether any changes are required.

The Lead Software Engineer shall notify team members of any changes to this plan at regular project meetings.